

Static Members and Methods

Static members belong to the **class itself** rather than any specific object. There is only one shared copy of a static variable in memory, regardless of how many objects you create.

Analogy: Think of a company office. Every employee (**Object**) has their own personal desk (**Instance Variable**). However, there is only one shared office Coffee Machine (**Static Variable**). You don't need to talk to a specific employee to read the company's general HR policy (**Static Method**).

1. Static Variables

- **What they do:** Create a single variable shared by every object of the class. If one object changes it, the change is reflected in all others.
- **Role:** Used for data that should be common to the entire group, like a counter.

2. Static Methods

- **What they do:** These can be called using the class name without creating an object. They can **only** access other static members.
- **Role:** Used for utility functions that don't need data from a specific instance.

```
class Car {
public:
    int price;
    static int totalCars; // Static Variable (Shared counter)

    // Increments the shared counter every time a car is made
    Car() { totalCars++; }

    static void showTotal() { // Static Method (Called without a specific car)
        cout << "Total cars made: " << totalCars;
    }
};

int Car::totalCars = 0; // Initialize static member outside the class

// Usage:
Car::showTotal(); // Outputs: 2
```

Under the Hood: Static variables are **not** stored inside the object's memory layout. The **sizeof(Car)** will only count the int **price**, ignoring **totalCars**. Static variables are stored globally in the program's Data Segment and exist for the entire lifetime of the program, even if no objects are created.

"Why can't a static method modify a regular class variable?" Because **static methods** do **not** receive the hidden **"this"** pointer. Since they don't have this, the compiler has absolutely no way of knowing which specific object's data you are trying to modify.